

1. Write a program that asks the user for their age and checks if they are eligible to vote (18 years and older). Print a message based on their eligibility.

```
age = int(input('Enter the age:'))

if age >= 18:
    print('Your Eligible to Vote')
else:
    print('Your Not Eligible to Vote')
```

Output:

Enter the age:22

Your Eligible to Vote

2. Write a program that takes a year as input and checks if it is a leap year or not. Print the result.

```
year = int(input('Enter the year : '))

if year % 400 == 0:
    print('It is a Leap Year')
elif year % 100 == 0:
    print('It is Not a Leap Year')
elif year % 4 == 0:
    print('It is a Leap Year')
else:
    print('It is a Not Leap Year ')
```

Output:

Enter the year : 2020

It is a Leap Year

3. Write a program that takes a number as input and checks if it is divisible by 5.

Print a message if it is divisible or not.

```
number = int(input('Enter a number: '))
```

```
if number % 5 == 0:
```

```
    print('Divisible by 5')
```

```
else:
```

```
    print('Not Divisible by 5')
```

Output:

Enter a number: 2

Not Divisible by 5

Enter a number: 150

Divisible by 5

4. Write a program that takes a character as input and checks if it is a vowel or consonant. Print the result.

```
ch = input('Enter a Character: ')
```

```
if ch in ('a','e','i','o','u'):
```

```
    print('Vowel')
```

```
else:
```

```
    print('Consonant')
```

Output:

Enter a Character: i

Vowel

Enter a Character: k

Consonant

5. Write a program that takes a password as input and checks if it is strong (at least 8 characters). Print a message based on the result.

```
password = input('Enter Password: ')
```

```
if len(password) >= 8:
```

```
    print('Strong Password')
```

```
else:
```

```
    print('Weak Password')
```

Output:

Enter Password: abcd@1234

Strong Password

Enter Password: xyz@123

Weak Password

6. Write a program that takes a grade (A, B, C, D, F) as input and uses match-case to print a message corresponding to the grade (e.g., "Excellent" for A, "Good" for B, etc.).

```
marks = (input('Enter the Grade: '))
```

```
match marks:
```

```
    case 'A':
```

```
        print('Excellent')
```

```
    case 'B':
```

```
        print('Good')
```

```
case 'C':  
    print('Average')  
case 'D':  
    print('Below Average')  
case 'F':  
    print('Fail')  
case _:  
    print('Invalid Grade')
```

Output:

Enter the Grade: A

Excellent

Enter the Grade: 50

Invalid Grade

7. Write a program that takes a traffic light color (Red, Yellow, Green) as input and uses match-case to print the corresponding action (e.g., "Stop" for Red, "Wait" for Yellow, etc.).

```
light = input('Enter the Color: ')
```

```
match light:
```

```
    case 'Red':
```

```
        print('Stop')
```

```
    case 'Yellow':
```

```
        print('Wait')
```

```
    case 'Green':
```

```
        print('Go')
```

```
    case _:
```

```
        print('Invalid Color')
```

Output:

Enter the Color: Yellow

Wait

8. Write a program that takes a number as input and uses a for loop to calculate its factorial. Print the result.

```
n = int(input('Enter a Number: '))
```

```
fact = 1
```

```
for i in range(1, n+1):
```

```
    fact = fact*i
```

```
print(fact)
```

Output:

Enter a Number: 4

24

Enter a Number: 7

5040

9. Write a program that uses a for loop to print all even numbers between 1 and 20.

```
for i in range(1, 21):
```

```
    if i % 2 == 0:
```

```
        print(i)
```

Output:

2

4

6

8

10

12

14

16

18

20

10. Write a program that uses a while loop to create a countdown timer from 10 to 0.

Print each number.

```
timer = 10
```

```
print("Starting countdown:")
```

```
while timer >= 0:
```

```
    print(timer)
```

```
    timer -= 1
```

Output:

Starting countdown:

10

9

8

7

6

5

4

3

2

1

0

11. Write a program that uses a for loop to count the number of vowels in a given string. Print the count.

```
str = input('Enter :')  
count = 0  
  
for ch in str:  
    if ch in 'a,e,i,o,u':  
        count+=1  
print(count)
```

Output:

Enter: apple

2

12. Write a program that uses a for loop to print numbers from 1 to 10, but skips printing the number 5 (use continue) and stops the loop if the number 8 is reached (use break).

```
for i in range(1,11):  
    if i == 5:  
        continue  
    if i == 8:  
        break  
    print(i)
```

Output:

1

2

3

4

6

7
